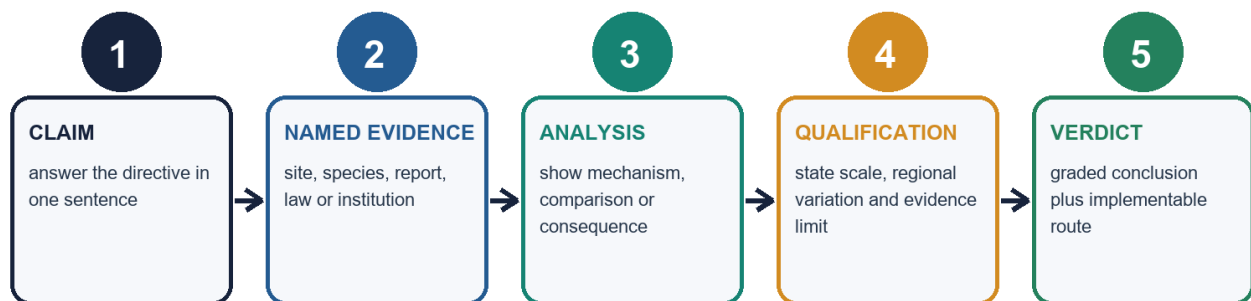


SOLVED PRACTICE WORKBOOK

Geography 23 - India Subalpine-Alpine Belt: Premium Solved Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

UPSC Mains Spine: Claim to Qualified Verdict



High-altitude answers score when they separate boundaries, regions, evidence types and governance stages

Original deterministic study schematic; not to scale.

Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

Workbook rules

- Objective answers rotate continuously **A -> B -> C -> D** from Q1 through Q32.
- Authentic PYQ option order is retained; PYQs are placed at rotation-compatible positions.
- Older objective keys unavailable in the local official-key holdings remain **INFERRED ANSWER - NOT OFFICIALLY VERIFIED**, with confidence and elimination.
- Every Mains solution follows **claim -> named evidence/example -> analysis -> qualification -> verdict**.
- Every solved Mains answer ends with **Why this earns marks**.

A. Solved Prelims PYQs, hard MCQs and remedial MCQs

Q1. UPSC Prelims 2020, GS-I, Q77 - official key unavailable locally

Which of the following are the most likely places to find the musk deer in its natural habitat?

1. Askot Wildlife Sanctuary
2. Gangotri National Park
3. Kishanpur Wildlife Sanctuary
4. Manas National Park

A. 1 and 2 only B. 2 and 3 only C. 3 and 4 only D. 1 and 4 only

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: A. Confidence: high.

Elimination: Askot and Gangotri are high-Himalayan protected landscapes suitable for musk-deer forest-subalpine habitat. Kishanpur is a Terai protected area, while Manas is primarily a foothill/alluvial and subtropical landscape rather than the paired high-Himalayan habitat tested.

Q2. Original - boundary firewall

Which statement is most accurate?

A. Treeline is the lower end of a glacier. B. Forest line, treeline, species line, snowline and ELA answer different ecological or glaciological questions. C. Timberline always has one universally accepted definition. D. Snowline is fixed by altitude alone.

Answer: B.

Explanation: Forest continuity, upright-tree form, highest species occurrence, persistent snow and glacier annual balance are distinct.

Q3. UPSC Prelims 2022, GS-I, Q45 - stem verified; official key unavailable locally

With reference to “Gucchi” sometimes mentioned in the news, consider the following statements:

1. It is a fungus.
2. It grows in some Himalayan forest areas.
3. It is commercially cultivated in the Himalayan foothills of north-eastern India.

Which of the statements given above is/are correct?

A. 1 only B. 3 only C. 1 and 2 D. 2 and 3

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: C. Confidence: high.

Elimination: Gucci refers to morel mushrooms, a fungus collected in Himalayan forest landscapes. The statement of established commercial cultivation in the north-eastern Himalayan foothills is not supported.

Q4. UPSC Prelims 2019, GS-I, Q18 - official scan verified; key unavailable locally

Which one of the following National Parks lies completely in the temperate alpine zone?

A. Manas National Park B. Namdapha National Park C. Neora Valley National Park D. Valley of Flowers National Park

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: D. Confidence: high.

Elimination: Manas is a foothill/alluvial landscape. Namdapha spans a very large tropical-to-alpine gradient. Neora Valley spans montane belts. Valley of Flowers is the high-Himalayan option identified with the temperate-alpine setting. The PYQ phrase should not erase the internal subalpine-alpine distinctions taught here.

Q5. UPSC Prelims 2019, GS-I, Q38 - official scan verified; key unavailable locally

Consider the following pairs:

1. Bandarpunch - Yamuna
2. Bara Shigri - Chenab
3. Milam - Mandakini
4. Siachen - Nubra
5. Zemu - Manas

Which pairs are correctly matched?

A. 1, 2 and 4 only B. 2 and 3 only C. 3 only D. 1, 2 and 3

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: A. Confidence: high.

Elimination: Bandarpunch-Yamuna, Bara Shigri-Chenab and Siachen-Nubra are correct. Milam is associated with Gori Ganga, not Mandakini; Zemu is associated with the Teesta system, not Manas.

Q6. Original - relative zonation

Which sequence is the safest broad transect?

A. Alpine meadow -> closed forest -> nival -> shrub B. Upper temperate forest -> subalpine forest -> krummholz/scrub -> alpine meadow/turf -> scree/fellfield -> nival C. Nival -> tropical forest -> alpine meadow D. Subtropical chir -> glacier snout -> species line

Answer: B.

Explanation: It preserves relative order while allowing overlap, local inversions and disturbance openings.

Q7. Original - treeline concepts

Which observation most directly demonstrates forest-line movement rather than only species-line change?

A. One seedling occurs above adult forest. B. One mature tree is taller after a warm summer. C. Repeated mapping shows the upper boundary of substantially continuous forest shifting, with stand structure and age evidence. D. Regional snowline was higher in one dry season.

Answer: C.

Explanation: Forest-line movement requires stand-continuity evidence through time; an isolated seedling indicates recruitment/species occurrence.

Q8. UPSC Prelims 2019, GS-I, Q31 - official scan verified; key unavailable locally

There was growing awareness in India about the importance of Himalayan nettle (*Girardinia diversifolia*) because it is found to be a sustainable source of:

A. Anti-malarial drug B. Biodiesel C. Pulp for paper industry D. Textile fibre

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: D. Confidence: high.

Elimination: Himalayan nettle yields bast fibre used for textiles. The item does not prove that every harvest is sustainable; regeneration, tenure and extraction method still matter.

Q9. Original - west-east template

Which statement is most defensible?

A. Western and Eastern Himalayan alpine belts differ in moisture regime and biogeographic composition; neither has one uniform species order. B. Eastern Himalaya has no winter snow. C. Western Himalaya is uniformly dry. D. Monsoon and western disturbances affect every slope identically.

Answer: A.

Explanation: Longitudinal circulation, inner/outer position, aspect and biogeography produce distinct mosaics.

Q10. Original - alpine plant adaptations

Which statement correctly interprets alpine adaptation?

A. Every alpine plant produces antifreeze proteins. B. Cushion/dwarf form, clonal growth, storage, pubescence, pigments, rapid phenology and dormancy are alternative portfolios that vary among taxa. C. Seed dormancy is absent because the growing season is short. D. Pollination is exclusively wind-driven.

Answer: B.

Explanation: Adaptation is species- and site-specific; no universal checklist applies.

Q11. Original - fauna context

Which combination is correctly contextualised?

A. Red panda - western cold-desert plateau; kiang - humid bugyal B. Black-necked crane - every Himalayan conifer forest; musk deer - Thar scrub C. Musk deer - upper forest/subalpine mosaics; kiang - Trans-Himalayan plateau contexts D. Himalayan monal - proof of old-growth forest

Answer: C.

Explanation: The item preserves habitat and regional distinctions.

Q12. UPSC Prelims 2020, GS-I, Q96 - official key unavailable locally

Siachen Glacier is situated to the:

A. East of Aksai Chin B. East of Leh C. North of Gilgit D. North of Nubra Valley

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: D. Confidence: high.

Elimination: The reliable geographical relationship is north of Nubra Valley. It also helps separate Karakoram cold-desert/cryosphere geography from humid alpine-meadow examples.

Q13. Original - alpine soils

Which soil statement is most accurate?

A. Stable moist meadows may develop organic-rich horizons, while ridges, scree and avalanche tracks can have thin or young soils. B. Every alpine soil is a deep podzol. C. Permafrost is universal across the Indian alpine belt. D. Meadow soil carbon can be inferred from forest cover.

Answer: A.

Explanation: Soil depth, moisture and carbon vary with landform, stability, snow and disturbance.

Q14. Original - snow insulation

Persistent snow can:

A. only damage vegetation; B. insulate soil and supply meltwater, yet also delay exposure and shorten growth where it persists late; C. determine the treeline without wind or soil; D. prove annual glacier mass gain.

Answer: B.

Explanation: Snow has opposing effects depending on timing, depth, density and site.

Q15. Original - ELA

Which definition is correct?

A. ELA is the highest tree species occurrence. B. ELA is the glacier snout. C. ELA is the glacier-specific annual balance line where accumulation equals ablation. D. ELA is the forest-line contour.

Answer: C.

Explanation: It is a glaciological budget boundary for a stated period.

Q16. Original - meadow debate

Which management approach is most defensible?

A. Afforest every treeless patch. B. Assume every opening is pristine. C. Ban all research because history is uncertain. D. Diagnose current flora, soil, snow/topography, land-use history, palaeoecology and imagery before classifying and restoring.

Answer: D.

Explanation: Natural, disturbance-maintained, secondary and degraded meadows require different responses.

Q17. Original - pastoralism

Which statement best captures grazing ecology?

A. Grazing can maintain heterogeneity or cause degradation depending on timing, intensity, herd composition, site and institutions. B. Grazing is always beneficial. C. Grazing is always destructive. D. Blanket exclusion never displaces pressure.

Answer: A.

Explanation: Thresholds and governance determine outcome.

Q18. UPSC Prelims 2021, GS-I, Q62 - official key unavailable locally

Consider the following statements:

1. The amount of water in the rivers and lakes is more than the amount of groundwater.
2. The amount of water in polar ice caps and glaciers is more than the amount of groundwater.

Which of the statements given above is/are correct?

A. 1 only B. 2 only C. Both 1 and 2 D. Neither 1 nor 2

INFERRED ANSWER - NOT OFFICIALLY VERIFIED: B. Confidence: high.

Elimination: Rivers/lakes are a smaller global water store than groundwater, while polar ice caps/glaciers store more. Global stock must not be confused with accessible annual basin flow.

Q19. Original - FRA

Which statement is correct?

A. FRA contains no relevance for pastoral communities. B. Recognition of a right automatically proves sustainable ecological use. C. FRA includes grazing and traditional seasonal resource access for nomadic/pastoral communities, while recognition, operation and ecological outcomes remain distinct. D. FRA converts every alpine meadow into private property.

Answer: C.

Explanation: The Act's right categories and implementation/ecological outcomes must be separated.

Q20. Original - medicinal plants and ABS

Which statement is most accurate?

A. Publishing rare-plant micro-locations improves conservation. B. Cultivation automatically makes wild harvest sustainable. C. A People's Biodiversity Register is a commercial extraction licence. D. Conservation should combine in situ stewardship, cultivation/propagation where suitable, traceability and applicable access-benefit sharing.

Answer: D.

Explanation: It links ecology, livelihoods and governance while protecting sensitive information.

Q21. Original - ecosystem services

Which claim is safest?

A. Alpine meadow can support forage, biodiversity, soil carbon, pollination, water regulation and cultural value, but cannot prevent every flood, landslide or avalanche. B. Meadow always increases annual basin flow. C. Shrub expansion always increases total ecosystem carbon. D. Tourism has no ecological cost below a fixed elevation.

Answer: A.

Explanation: Services are real but process- and scale-bounded.

Q22. Original - climate evidence

Which statement follows strong evidence grammar?

A. One warm season proves climate change. B. A claim should name site, slope, period, variable and method, then separate observation, mechanism, attribution and projection. C. A model projection is an observed outcome. D. One seedling proves forest-line migration.

Answer: B.

Explanation: It prevents spatial and causal overclaiming.

Q23. Original - treeline paradox

Which outcome is possible under warming?

A. Only upward treeline advance. B. Only retreat. C. Recruitment or infilling may increase, but wind, snow, moisture stress, shallow soil, grazing, avalanches and seed limitation can produce stasis or reversal. D. Local controls become irrelevant.

Answer: C.

Explanation: Warming changes the thermal opportunity, not every local filter.

Q24. Original - data literacy

Why can FSI forest-cover data not directly measure alpine meadow condition?

A. FSI uses no remote sensing. B. Meadows are legally outside India. C. Forest cover is always field inventory. D. Canopy-density mapping does not by itself measure meadow composition, soil, grazing, shrub cover or erosion.

Answer: D.

Explanation: Match the metric to the question.

Q25. Original - governance stack

Which sequence is analytically correct?

A. Legal status/right/clearance/plan/funding/compliance/ecological outcome are separate stages or categories. B. World Heritage status automatically sets a visitor quota. C. CAMPA funding proves native recovery. D. A clearance proves absence of impact.

Answer: A.

Explanation: Governance analysis fails when documents and outcomes are collapsed.

Q26. Original - restoration hierarchy

What should usually precede planting?

A. A tree-count target. B. Avoiding conversion, diagnosing the ecosystem and repairing drainage/soil while protecting natural regeneration. C. Replacing native turf with fast-growing exotics. D. Ignoring rights and access.

Answer: B.

Explanation: Avoidance and process repair are higher in the hierarchy than planting.

Q27. Original - regional cases

Which comparison is correct?

A. Hemis/Changthang are humid eastern bugyals. B. Valley of Flowers and Changthang have identical hydrology. C. Khangchendzonga illustrates an eastern full-altitudinal and krummholz-rich landscape, while Hemis/Changthang are Trans-Himalayan cold-desert/plateau contexts. D. Nanda Devi is a tropical coastal park.

Answer: C.

Explanation: It preserves east-west and alpine-cold-desert distinctions.

Q28. Original - monitoring

Which dashboard is most complete?

A. Tree height alone B. Visitor revenue alone C. NDVI alone D. Treeline demography, shrub cover, snow, soil, composition, phenology, pressure, wildlife, water, rights, livelihoods and compliance

Answer: D.

Explanation: Ecotone condition is multidimensional.

Q29. Remedial - wrong-site afforestation

Why can planting trees in a natural alpine meadow be harmful?

A. It can replace open-habitat biota and alter snow, albedo, hydrology and grazing/wildlife use. B. Trees are always legally prohibited above 1,000 m. C. Meadow has no soil. D. Every sapling causes a GLOF.

Answer: A.

Explanation: The objection is ecological mismatch, not opposition to all restoration.

Q30. Remedial - blanket grazing exclusion

Which is the strongest caution?

A. Exclusion always improves every meadow. B. Exclusion can displace pressure, break seasonal institutions and encourage shrub change in some history-shaped sites; site thresholds are still required. C. Rights remove all conservation duties. D. Herders never monitor snow or forage.

Answer: B.

Explanation: Social and ecological feedbacks must be tracked.

Q31. Remedial - remote-sensing greening

What does a greening signal require before ecological interpretation?

A. Immediate plantation target B. No field work C. Season/snow/cloud checks plus field evidence on shrubs, native composition, biomass, soil and pressure D. Assumption that greening equals biodiversity gain

Answer: C.

Explanation: Greening is a signal, not a cause or complete condition measure.

Q32. Remedial - permafrost claim

Which statement is correct?

A. Permafrost occurs uniformly beneath every Himalayan meadow. B. Any winter-frozen soil is permafrost. C. Permafrost can be inferred from altitude alone. D. Permafrost requires site evidence and varies with climate, aspect, ground cover, substrate and terrain.

Answer: D.

Explanation: Seasonal frost and permafrost are distinct.

B. Solved Mains PYQs

PYQ 1 - UPSC Mains 2019, GS-I, Q15

Question: How can the mountain ecosystem be restored from the negative impact of development initiatives and tourism? (15 marks, 250 words)

Demand: "How can" requires an implementable hierarchy. "Mountain ecosystem" requires slope, water, forest-meadow ecotones, biodiversity, hazard and community integration.

Model solution

- **Claim:** Mountain restoration must begin with avoidance because thin soils, steep relief, slow succession and vertically connected habitats make post-impact repair incomplete.
- **Named evidence/example:** UNESCO's Nanda Devi-Valley of Flowers profile identifies alpine meadows, glaciers, threatened species, regulated access and buffer-zone development/tourism risks. In the Kedarnath-upper Himalayan context, routes, camps, roads, drainage and extreme-weather exposure overlap.
- **Analysis:** First, map no-go/refugia zones around natural meadows, snowbeds, wetlands, unstable slopes, avalanche paths and wildlife/pastoral corridors. Second, replace isolated project appraisal with cumulative basin/corridor assessment. Third, enforce hill-road drainage, spoil disposal, slope audits and native-site restoration. Fourth, regulate visitors through seasonal permits, route/camp zoning, sanitation, waste accountability, water limits and weather closures. Fifth, co-manage grazing/seasonal access with rights holders using site thresholds. Sixth, control invasives and protect medicinal-plant populations. Finally, monitor soil erosion, native composition, snow/water timing, wildlife, visitor pressure, compliance and livelihoods.
- **Qualification:** Vegetation can reduce shallow erosion and protect soil, but cannot prevent every deep landslide, avalanche or extreme flood. Plantation cannot quickly recreate old meadow/forest soil and interaction networks.
- **Verdict:** Restoration is an **avoid -> regulate -> repair -> co-govern -> monitor** programme, not a compensatory tree-count exercise.

Why this earns marks: It directly answers “how,” uses a named protected landscape, integrates engineering, ecology, tourism and rights, states physical limits and gives measurable implementation.

PYQ 2 - UPSC Mains 2020, GS-I, Q6

Question: How will the melting of Himalayan glaciers have a far-reaching impact on the water resources of India? (10 marks, 150 words)

Demand: Explain multiple water-resource pathways without claiming that all rivers immediately dry.

Model solution

- **Claim:** Glacier loss matters mainly by changing the timing, reliability and hazard profile of glacierised headwaters.
- **Named evidence/example:** Gangotri-Gomukh is the Bhagirathi headstream cue; the glacier-river PYQ links Bara Shigri with Chenab and Zemu with the Teesta system, illustrating basin specificity.
- **Analysis:** Sustained negative mass balance can initially add melt locally, but declining ice storage later weakens lean-season and drought-year buffering. Hydropower operation, irrigation scheduling, sediment management, ecosystem flows and storage design face greater uncertainty. Retreat can also leave or enlarge moraine-dammed lakes and add GLOF pathways. Alpine snowbeds, wetlands and soils further mediate the seasonal pulse.
- **Qualification:** Monsoon rain supplies much annual flow in many basins; snow, groundwater, debris cover, elevation and glacier fraction vary. Snowline, ELA and glacier snout are different indicators.
- **Verdict:** The strongest conclusion is reduced seasonal reliability and higher planning complexity, requiring basin-specific monitoring and risk-aware infrastructure.

Why this earns marks: It explains phased hydrology, gives named basin evidence, distinguishes glacier/snow metrics and avoids the “all Himalayan rivers vanish” exaggeration.

PYQ 3 - UPSC Mains 2021, GS-I, Q4

Question: Differentiate the causes of landslides in the Himalayan region and Western Ghats. (10 marks, 150 words)

Demand: Compare causes; do not list generic triggers without geological and climatic differentiation.

Model solution

- **Claim:** Both regions experience rain-triggered slope failure, but the young tectonic Himalaya and old dissected Western Ghats differ in structure, cryosphere influence and development context.
- **Named evidence/example:** Himalayan slopes combine active tectonics, fractured/folded rocks, high relief, river incision, glacial/periglacial debris, snow/avalanche processes and road/hydropower cutting. Western Ghats failures commonly involve deeply weathered regolith/laterite, escarpment relief and intense monsoon rain, with quarrying and slope modification as local amplifiers.
- **Analysis:** In the Himalaya, earthquakes, thrust-related weakness, glacial/moraine material, freeze-thaw and extreme rain/snow interact with rapidly incised valleys. In the Ghats, prolonged chemical weathering and saturation of thick soil/regolith are more prominent. Drainage obstruction, deforestation and construction intensify both.
- **Qualification:** No region has one landslide mechanism; site lithology, slope, antecedent moisture and engineering decide failure type.
- **Verdict:** Risk mapping must be process-specific: tectonic-cryosphere-corridor integration in the Himalaya, and weathered-mantle-rainfall-drainage/quarry focus in the Ghats.

Why this earns marks: It differentiates rather than narrates, names geology, climate and development controls, and preserves site-level variation.

PYQ 4 - UPSC Mains 2020, GS-I, Q17

Question: Examine the status of forest resources of India and its resultant impact on climate change. (15 marks, 250 words)

Demand: Examine needs evidence, interpretation and counterpoint. The alpine link is forest-cover and treeline data literacy.

Model solution

- **Claim:** India's forest resource is nationally mapped, but canopy extent cannot alone reveal native composition, fragmentation, old growth, treeline dynamics or alpine-meadow condition.
- **Named evidence/example:** FSI's ISFR 2023 is the current official forest-cover baseline used in this package. Its canopy methodology can include diverse land covers meeting the threshold; it should not be read as a subalpine-forest quality or meadow-health survey.
- **Analysis:** Forests mitigate climate change through biomass and soil carbon and support water, slope and microclimate functions. Himalayan upper forests also provide seed sources and vertical corridors. Yet fragmentation, fire, grazing/browse imbalance, roads and poor regeneration can reduce resilience while canopy remains. Wrong-site plantation can harm natural meadows; assisted natural regeneration and protection of native mosaics can support mitigation and adaptation.
- **Qualification:** Forest cover, recorded forest area, carbon stock, native ecosystem and legal status are different. Greening may be shrubification, plantation or seasonal signal; one fire/snow season is not a trend.
- **Verdict:** Forest-climate policy should retain canopy monitoring but judge outcomes through native composition, age/regeneration, connectivity, soil carbon, hydrology, disturbance and rights.

Why this earns marks: It uses a named official baseline, exposes the metric's blind spot, applies it to the subalpine-alpine ecotone and balances mitigation value with ecological non-equivalence.

C. Original solved Mains practice

Original 10-marker - boundary distinctions

Question: Distinguish treeline, timberline, snowline and equilibrium-line altitude. Why is their separation important in Himalayan climate analysis? (10 marks, 150 words)

Model solution

- **Claim:** The four terms describe different biological, landscape and glaciological boundaries and therefore cannot be used as proxies for one another.
- **Named evidence/example:** Near Himalayan upper forests, forest continuity can end below isolated upright birch/fir or krummholz; by contrast, snowline marks persistent snow, while the ELA marks annual glacier accumulation-ablation balance. The glacier snout lies still lower on many valley glaciers.
- **Analysis:** Treeline concerns upright tree growth; timberline often denotes continuous forest or usable timber and must be defined; snowline depends on snow persistence and surface climate; ELA is glacier- and year-specific. Wind, aspect, grazing and avalanche can lower observed treeline without shifting ELA. A negative glacier balance can raise ELA and retreat the snout without immediate forest-line movement.
- **Qualification:** Terminology varies among authors, so scale, period and operational definition must be stated.
- **Verdict:** Separation prevents false climate attribution and enables correct monitoring of vegetation, snow and glacier change.

Why this earns marks: It defines each term, uses a slope/glacier example, explains analytical consequences and qualifies terminology.

Original 15-marker - pastoralism and meadow conservation

Question: "Grazing is both a driver of alpine-meadow degradation and a component of Himalayan cultural landscapes." Critically examine. (15 marks, 250 words)

Model solution

- **Claim:** The statement is valid only through a threshold lens: grazing outcome depends on meadow origin, season, intensity, herd composition, moisture, slope and institutions.
- **Named evidence/example:** Majid and Khullar describe summer use of alpine pastures and transhumance; FRA recognises grazing and seasonal access categories. Uttarakhand bugyals and Himachal/Kashmir high pastures illustrate diverse commons and protected-area contexts rather than one system.
- **Analysis:** Moderate, rotational use can maintain patch heterogeneity, recycle nutrients, support local breeds/livelihoods and restrain shrubs in some history-shaped openings. Conversely, early-season use on saturated soils, concentrated camps/routes, excessive stocking and loss of rotation can compact soil, expose bare ground, shift palatable species and damage wetlands. Blanket exclusion may displace pressure, erode knowledge and undermine livelihoods; blanket access can ignore ecological thresholds and wildlife needs.
- **Qualification:** Natural climatic meadow, grazing-maintained opening and degraded patch require different baselines. Rights recognition does not predetermine stocking level, and ungrazed enclosures can alter vegetation in ways that need interpretation.
- **Verdict:** Use rights-based co-management with mapped sensitive sites, seasonal calendars, negotiated thresholds, pressure-displacement tracking and joint ecological-livelihood monitoring.

Why this earns marks: It presents both mechanisms, names book/legal evidence, distinguishes meadow histories and ends with an implementable balanced model.

Original 20-marker - climate-resilient governance

Question: Design a climate-resilient conservation and restoration strategy for India's subalpine-alpine belt. (20 marks, 300 words)

Model solution

- **Claim:** Resilience requires protecting heterogeneous ecotones, snow-soil processes and livelihood institutions rather than assuming uniform upward vegetation migration or relying on plantations.
- **Named evidence/example:** GBPNHE's Dharma-Parvati project links land use/elevation with soil nutrients, microbial biomass and enzyme activity; UNESCO's Nanda Devi-Valley of Flowers and Khangchendzonga profiles demonstrate western and eastern high-altitude mosaics; WII's SPAI illustrates landscape-scale monitoring.
- **Analysis:**
 1. **Map correctly:** forest line, treeline, species line, shrub, meadow, snowbed, scree, wetland and corridor at sector/slope scale.
 2. **Avoid:** no-go natural meadows, refugia, avalanche/wetland processes and unstable slopes; prevent wrong-site afforestation and cumulative fragmentation.
 3. **Regulate:** site-season grazing, route/camp/waste/water limits, medicinal harvest, fire and infrastructure drainage/spoil.
 4. **Restore processes:** repair hydrology and soil, assist natural regeneration, use native meadow vegetation where diagnosed, control invasives and steward medicinal plants.
 5. **Co-govern:** integrate FRA/community rights, protected-area plans, BMC/PBR/ABS, NMPB support, CAMPA/GIM and tourism/disaster protocols while separating funding from outcome.
 6. **Monitor:** tree demography, shrub cover, snow/melt, soil carbon/moisture/microbes, composition, pollinators, grazing/visitors, wildlife, water, rights and compliance.
- **Qualification:** Projections guide risk screening, but local intervention needs field trends. Natural meadows, secondary openings and degraded patches are not interchangeable; restoration cannot instantly recreate old soil or rare assemblages.

- **Verdict:** The strategy is **refugia + connectivity + threshold use + process restoration + rights + long-term evidence**, differentiated west-to-east and from the Trans-Himalayan cold desert.

Why this earns marks: It uses three named evidence units, integrates ecology, climate, law and livelihoods, gives a sequenced implementation plan and states regional/data limitations.
